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Iatrogenic botulism outbreak associated with aesthetic procedures using injectable botulinum toxin products

The UK Health Security Agency (UKHSA) along with Medicines and Healthcare products Regulatory Agency (MHRA), local authorities and the NHS are investigating an ongoing outbreak of iatrogenic botulism in England.

From 4 June, 2025, to 6 August, 41 clinically confirmed cases of iatrogenic botulism had been reported to UKHSA from across 5 regions of England (North East, East Midlands, East of England, North West, Yorkshire and Humber). Thirty-two of the cases had been admitted to hospital with 8 requiring treatment in intensive care. All cases had reported having recent aesthetic procedures involving injections of botulinum toxin, with investigations showing that the majority of individuals received products that are unlicensed for use in the UK.

UKHSA and its partners issued an alert urging the public to take precautions when seeking aesthetic procedures and to seek medical advice if they experienced symptoms suggestive of botulism such as: difficulty swallowing; slurred speech; and breathing difficulties.

The public have been advised to ensure they are receiving products that are licensed for use in England and that practitioners are sufficiently trained to give procedures involving injectable botulinum toxin. Advice is available on the NHS website (https://www.nhs.uk/tests-and-treatments/cosmetic-procedures/advice/choosing-who-will-do-your-procedure/) to support the public in choosing who will undertake their cosmetic procedures.

Injectable botulinum toxin type products currently licensed in the UK for aesthetic procedures include Alluzience®, Azzalure®, Bocouture®, Botox®, Letybo®, Nuceiva® and Relfydess®.

Guidance is available through MHRA's website to support the public and practitioners identifying potentially fake licensed products and online retailers (https://fakemeds.campaign.gov.uk/spot-a-fake/) and to report unlicensed products (https://yellowcard.mhra.gov.uk/) that are in circulation.

Real-time syndromic surveillance in England: the first 25 years

The UK Health Security Agency (UKHSA) Field Services Real-time Syndromic Surveillance Team (ReSST) coordinates 5 national syndromic surveillance systems, out of which real-time surveillance outputs and epidemiological intelligence are delivered to support national UKHSA disease surveillance programmes and national/local incident response.

Weekly updates from the following 5 syndromic surveillance systems are published on GOV.UK (1) covering different categories of English syndromic data:

- the 'remote health advice' system (https://www.gov.uk/government/publications/remote-health-advice-weekly-bulletins-for-2025) (monitoring patterns in NHS 111 (https://111.nhs.uk/?gad_source=1&gad_campaignid=22485231558&gclid=CjwKCAjw2brFBhBOEiwAVJX5GMtDFiUZXrRloBkogewRQtZiIwcOvgjCUYSs-hNQ_R6m57aPRMkdchoCGQoQAvD_BwE) phone calls and online activity);
- the 'GP in-hours' system (https://www.gov.uk/government/collections/syndromic-surveillance-systems-and-analyses#gp-in-hours-syndromic-surveillance-system) (monitoring the number of patients consulting with GPs during regular surgery hours for known clinical indicators)
- the 'GP out-of-hours' system (https://www.gov.uk/government/collections/syndromic-surveillance-systems-and-analyses#gp-out-of-hours-syndromic-surveillance-system) (monitoring unscheduled GP visits, such as during evenings, weekends and bank holidays)
- the 'emergency department' system (https://www.gov.uk/government/collections/syndromic-surveillance-systems-and-analyses#emergency-department-syndromic-surveillance-system) (monitoring attendances across a network of emergency departments throughout the country), and
- the 'national ambulance' system (https://www.gov.uk/government/collections/syndromic-surveillance-systems-and-analyses#national-ambulance-syndromic-surveillance-system) (which monitors calls made to the 10 ambulance trusts operating in England)

The genesis of syndromic surveillance in England can be traced back to the late 1990s (in the Public Health Laboratory Service) when a small exploratory and opportunistic pilot project was undertaken to collect data from the fledgling NHS telephone health service (NHS Direct, now NHS 111). The pilot project demonstrated that monitoring daily NHS Direct 'cold or flu' calls could provide earlier warning of seasonal influenza activity than could the GP- and laboratory-based influenza surveillance systems that were operational at the time (2). The success of this pilot initiated a programme of development of real-time syndromic surveillance over the following two decades (3): from a single-region, single-indicator pilot (using manual data transfer methods) to a suite of systems that now monitors a

wide range of syndromes from acute respiratory illnesses to diarrhoea to cardiac conditions. It is widely used in routine public health surveillance, monitoring seasonal respiratory disease and incidents such as the COVID-19 pandemic.

The post-pilot development and expansion of the English syndromic surveillance programme was in part due to several key events and drivers that highlighted the need for real-time epidemiological intelligence. The 1990s saw the re-emergence of a new public health threat: bioterrorism. Overseas, the US 2001 ‘September 11’ attacks and ‘anthrax mail’ incidents highlighted the need for surveillance systems to monitor the health of the population to identify unusual healthcare presentations in near real-time. This form of surveillance (termed ‘syndromic’) captured health data including the presentation of chief complaints, symptoms or provisional diagnoses presenting to health care providers, rather than laboratory confirmed cases of disease (4).

In England, during the 2000s, major public health incidents such as the 2003 European heatwave, the 2005 Buncefield Oil Depot explosion and fire, the 2009 influenza pandemic and the 2010 Icelandic volcanic eruption and ash cloud provided the opportunity for syndromic surveillance to showcase the value of real-time intelligence, informing incident response teams on the impact (or lack of impact) of each incident on the health of the population.

A key driver for the development of the syndromic surveillance programme was the London 2012 Olympic and Paralympic Games. The Games provided a major challenge to the then Health Protection Agency to prepare health protection systems to cope with the increased demands and threats of this major mass gathering. For syndromic surveillance, this required expansion of the programme by developing new systems (adding real-time monitoring of emergency department attendances and GP out-of-hours contacts), developing and incorporating bespoke statistical exceedance systems and risk assessments into the daily surveillance routine, and expanding the team to cope with the increased demands of real-time (daily) surveillance (5,6). The legacy of the 2012 Games was the continuation of the expanded and enhanced national syndromic surveillance programme (7). Since 2012, syndromic surveillance has been an integral component of routine surveillance of respiratory and gastrointestinal infections, environmental incidents (eg heatwaves) and national emergencies such as the COVID-19 pandemic in England.

A key factor in the success of the programme has been the technical advancements in data and system infrastructure, data and epidemiological analysis of data. However, an often-overlooked critical component of this success is the importance of the ‘human’ aspect of this work (8) . Experience is critical for: understanding and interpreting syndromic data; assessing potential new syndromic data feeds and developing new systems; and for maintaining relationships with syndromic data providers and strengthening partnerships with academic units to maximise the

usefulness of syndromic data for health research. Recent reflections on the 25-year history of the programme have also highlighted the importance of having senior 'champions' in the organisation to support, steer and safeguard the programme (9).

The advent of new technologies will improve data access, automation, storage, interrogation and security, which will facilitate more innovative developments to enhance the power and usefulness of the data. The current era of artificial intelligence and machine learning could also potentially facilitate the opening of new avenues for innovative approaches to fully explore the maximum utility of the syndromic systems. However, two key targets of our future UKHSA syndromic surveillance programme are: firstly, sharing the experience of the UKHSA's ReSST internationally to ensure that syndromic surveillance continues to be an international standard for epidemiological surveillance; and, secondly, training the next generation of epidemiologists and scientists to carry syndromic surveillance forward through the next quarter of a century, and beyond.

> This article was written by Alex Elliot, Helen Hughes and Dan Todkill on behalf of the Real-time Syndromic Surveillance Team in the UKHSA Field Services Division.

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